

- (c) Part of the stellar system
- (d) Tailles Comet

I.A.S. (Pre) 1995

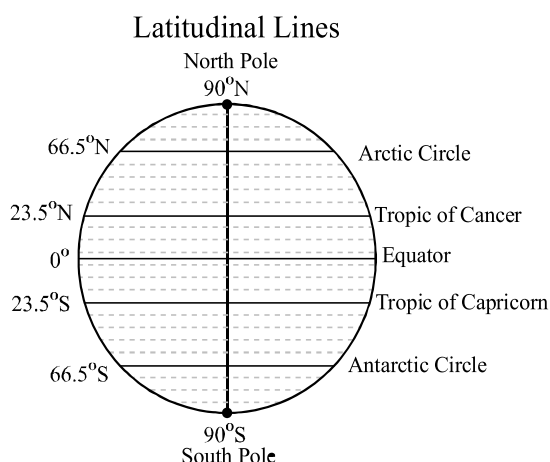
Ans. (b)

A meteor is a bright streak of light in the sky also called a shooting star or a falling star but actually, it is not a star but it is just a small piece of rock and minerals. It starts glowing because when it enters the Earth's atmosphere with a very high velocity it burns due to heat caused by air friction. Most of the meteors burn to ashes before reaching the surface of the Earth.

The Earth

i. Latitudes

*Latitude is the angular distance of a point on the earth's surface, measured in degrees from the center of the earth. *It is parallel to the line of the equator, which lies midway between the poles. *The latitude angle is denoted by the Greek lowercase letter Phi (ϕ). 0° latitude is also known as the **equator**. It is halfway between the North Pole and the South Pole. The equator thus divides the planet into the Northern and Southern Hemisphere.



The earth is widest at the equator. All other latitudes are smaller than the equator. *The North Pole has a latitude of 90°N and the South Pole has a latitude of 90°S . In each hemisphere, latitudes are counted from 0° to the poles. Hence, altogether there are 179 latitudes (including the equator). Besides this, there are 179 real circle latitudes as the 90°N and 90°S latitudes are points on the earth's surface. *The 23.5°N latitude in the Northern Hemisphere is known as the **Tropic of Cancer** whereas the 66.5°N latitude in the Northern Hemisphere is known as **Arctic Circle**. *Similarly,

the 23.5°S latitude in the southern Hemisphere is known as the **Tropic of Capricorn** and 66.5° south latitude in the Southern Hemisphere is known as the **Antarctic Circle**.

*The 0° latitude (equator) and 0° longitude (Prime Meridian) intersect each other in the Gulf of Guinea located on the western coast of Africa. The Gulf is located in the Atlantic Ocean. The north and south of any place on a globe or a map is determined by latitudes. The latitudinal value in degrees of any place is directly proportional to its extent in the north of the equator for example - Beijing (China) - $39^\circ55'\text{N}$, New Delhi - $28^\circ38'\text{N}$, New York (USA) - $40^\circ45'\text{N}$ and Rome (Italy) - $41^\circ54'\text{N}$. Hence among these Rome is the most northern city.

1. **Correct Latitudinal position of a place that could be possible is-**

- (a) 91°North
- (b) 45°East
- (c) 45°South
- (d) 91°West

R.A.S. / R.T.S. (Pre) 1999

Ans. (c)

Latitudes in the Greek language are generally denoted by the letter ϕ . It shows the places towards North and South of the Equator. 90° Northern latitude is called the North Pole while 90° Southern latitude is called the South Pole which are just points not the circles, but other latitudes on both sides of the Equator are imaginary circles. East and West of Prime Meridian are a total 360 longitudes. So, in the above options obviously 45°East and 91°West are representing longitudinal positions and 91°North has no meaning. Therefore 45°South represents the correct latitudinal position of a place.

2. **Which of the following is not a great circle on the globe?**

- (a) Equator
- (b) Prime meridian line
- (c) $60^\circ\text{East longitude}$
- (d) $60^\circ\text{North latitude}$

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Ans. (d)

A great circle is that line which divides a sphere into two hemispheres. Upon the Earth, all the meridians longitudes and the Equator are great circles. Except for the equator no other latitude is a great circle because they are smaller in comparison to the Equator.

3. **Zero degree latitude and zero degree longitude lies in the**

- (a) Atlantic Ocean
- (b) Arctic Ocean